

姓名: _____

学院: _____

学号: _____

上海科技大学

2025-2026 学年第 1 学期本科生期中试卷

开课单位: 信息学院

授课教师: 文鼎柱&吴幼龙

考试科目: 《面向信息科学的概率论与数理统计》

课程代码: SI 140A

考试时间: 2025 年 11 月 18 日 10 点 15 分-11 点 55 分

考试成绩录入表:

题目	1	2	3	4	5	6	7	8	总分
计分									
复核									

评卷人签名:

复核人签名:

日期:

日期:

编写说明:

1. 要求评卷人和复核人不能是同一人。
2. 试卷内页和答题纸编排格式由各学院和出题教师根据实际需要自定, 每页须按顺序标注页码(除封面外), 要求排版清晰、美观, 便于在页面左侧装订。为方便印刷归档, 建议使用 A4 双面印刷(学校有印刷一体机提供)。
3. 主考教师编写试卷时尽可能保证试题科学、准确、合理, 如考试过程中发现试题有误, 主考教师需负责现场解释, 此类情况学校将作为教学评估记录的一部分。

Probability & Statistics

Name (Print): _____

Fall 2025

Midterm

2025/11/18

Time Limit: 100 Minutes

Advisor Name _____

This exam contains 10 pages (including this cover page) and 8 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

Try to answer as many problems as you can. The following rules apply:

- **Mysterious or unsupported answers will not receive full credit.** A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit.

Do not write in the table to the right.

Problem	Points	Score
1	10	
2	10	
3	15	
4	10	
5	15	
6	15	
7	10	
8	15	
Total:	100	

1. (10 points) Let events A and B satisfy $P(A) = 0.6$, $P(B) = 0.4$, and $P(A \cap B) = 0.2$.
 - (a) (5 points) Calculate the probability of the symmetric difference $P(A \triangle B)$.
(Hint: $A \triangle B = (A \cap B^c) \cup (A^c \cap B)$.)
 - (b) (5 points) Verify that your result satisfies the identity $P(A \triangle B) = P(A) + P(B) - 2P(A \cap B)$.

2. (10 points) hat contains 100 coins, where at least 99 are fair, but there may be one that is double-headed (always landing Heads); if there is no such coin, then all 100 are fair. Let D be the event that there is such a coin, and suppose that $P(D) = 1/2$. A coin is chosen uniformly at random. The chosen coin is flipped 7 times, and it lands Heads all 7 times.
- (a) (5 points) Given this information, what is the probability that one of the coins is double-headed?
 - (b) (5 points) Given this information, what is the probability that the chosen coin is double-headed?

3. (15 points) Alice flips a fair coin n times and Bob flips another fair coin $n + 1$ times, resulting in independent $X \sim \text{Bin}(n, \frac{1}{2})$ and $Y \sim \text{Bin}(n + 1, \frac{1}{2})$.
- (a) (8 points) Show that $P(X < Y) = P(n - X < n + 1 - Y)$.
- (b) (7 points) Compute $P(X < Y)$.
(Hint: Use (a) and the fact that X and Y are integer-valued.)

4. (10 points) A gambler repeatedly plays a game where in each round, he wins a dollar with probability $\frac{1}{3}$ and loses a dollar with probability $\frac{2}{3}$. His strategy is “quit when he is ahead by \$2”, though some suspect he is a gambling addict anyway. Suppose that he starts with a million dollars. Show that the probability that he’ll ever be ahead by \$2 is less than $\frac{1}{4}$.

5. (15 points) Let X and Y be independent geometric random variables, where X has parameter p and Y has parameter q .
- (a) (5 points) What is the probability that $X = Y$?
 - (b) (5 points) What is $E[\min(X, Y)]$?
 - (c) (5 points) What is $E[\max(X, Y)]$?
(Hint: Use the relation $\max(X, Y) = X + Y - \min(X, Y)$)

6. (15 points)

- (a) (5 points) Let W be a bounded non-negative integer-valued random variable. If for all integer $k \geq 0$,

$$E \left[\binom{W}{k} \right] \approx \frac{\lambda^k}{k!}$$

Show that

$$P(W = k) \approx e^{-\lambda} \frac{\lambda^k}{k!}$$

- (b) (5 points) Consider independent trials in which each trial is equally likely to result in any of r possible outcomes, where r is very large. Let X denote the number of outcomes that have occurred at least m times in the first n trials. Find the approximate distribution of X .
- (c) (5 points) There are n people in a room. Assume each person's birthday is equally likely to be any of the 365 days of the year (we exclude February 29), and that people's birthdays are independent (we assume there are no twins in the room). When the probability that three or more people in the group have the same birthday is $1/2$, find n .

7. (10 points) Let $X \sim \text{Pois}(\lambda)$, where λ is fixed but unknown. Let $\theta = e^{-3\lambda}$, and suppose that we are interested in estimating θ based on the data. Since X is what we observe, our estimator is a function of X , call it $g(X)$. The *bias* of the estimator $g(X)$ is defined to be $E(g(X)) - \theta$, i.e., how far off the estimate is on average; the estimator is *unbiased* if its bias is 0.
- (a) (5 points) For estimating λ , the r.v. X itself is an unbiased estimator. Compute the bias of the estimator $T = e^{-3X}$. Is it unbiased for estimating θ ?
- (b) (5 points) Show that $g(X) = (-2)^X$ is an unbiased estimator for θ .

8. (15 points) A department consists of w professors and b students. A team of n people is formed by choosing members randomly without replacement from the department. Let the random variable X be the number of professors on the team. We are given that X follows a Hypergeometric distribution, and its mean is $E[X] = n \frac{w}{w+b}$.
- (a) (7 points) To enhance collaboration, members of the team are to be paired up. Find the expected number of distinct pairs consisting of one professor and one student on the team.
- (b) (8 points) Using your result from part (a) and the given mean $E[X]$, find the variance of the number of professors on the team, $\text{Var}(X)$.

Table 1: Bayes' Rule

	Y discrete	Y continuous
X discrete	$P(Y = y X = x) = \frac{P(X = x Y = y)P(Y = y)}{P(X = x)}$	$f_Y(y X = x) = \frac{P(X = x Y = y)f_Y(y)}{P(X = x)}$
X continuous	$P(Y = y X = x) = \frac{f_X(x Y = y)P(Y = y)}{f_X(x)}$	$f_{Y X}(y x) = \frac{f_{X Y}(x y)f_Y(y)}{f_X(x)}$

Table 2: Law of Total Probability (LOTP)

	Y discrete	Y continuous
X discrete	$P(X = x) = \sum_y P(X = x Y = y)P(Y = y)$	$P(X = x) = \int_{-\infty}^{\infty} P(X = x Y = y)f_Y(y) dy$
X continuous	$f_X(x) = \sum_y f_X(x Y = y)P(Y = y)$	$f_X(x) = \int_{-\infty}^{\infty} f_{X Y}(x y)f_Y(y) dy$

Name	Param.	PMF or PDF	Mean	Variance
Bernoulli	p	$P(X = 1) = p, P(X = 0) = q$	p	pq
Binomial	n, p	$\binom{n}{k} p^k q^{n-k}, k \in \{0, 1, \dots, n\}$	np	npq
FS	p	$pq^{k-1}, k \in \{1, 2, \dots\}$	$1/p$	q/p^2
Geom	p	$pq^k, k \in \{0, 1, 2, \dots\}$	q/p	q/p^2
NBinom	r, p	$\binom{n+r-1}{n} p^r q^n, n \in \{0, 1, 2, \dots\}$	rq/p	rq/p^2
HGeom	w, b, n	$\frac{\binom{w}{k} \binom{b}{n-k}}{\binom{w+b}{n}}, k \in \{0, \dots, n\}$	$\mu = \frac{nw}{w+b}$	$\left(\frac{w+b-n}{w+b-1}\right) n \frac{\mu}{n} \left(1 - \frac{\mu}{n}\right)$
Poisson	λ	$e^{-\lambda} \lambda^k / k!, k \geq 0$	λ	λ
Uniform	$a < b$	$\frac{1}{b-a}, x \in (a, b)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal	μ, σ^2	$\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$	μ	σ^2
Log-Normal	μ, σ^2	$\frac{1}{x\sigma\sqrt{2\pi}} e^{-(\log x - \mu)^2/(2\sigma^2)}$	$\theta = e^{\mu + \sigma^2/2}$	$\theta^2(e^{\sigma^2} - 1)$
Exponential	λ	$\lambda e^{-\lambda x}, x > 0$	$1/\lambda$	$1/\lambda^2$
Gamma	a, λ	$\frac{\lambda^a x^{a-1} e^{-\lambda x}}{\Gamma(a)}, x > 0$	a/λ	a/λ^2
Beta	a, b	$\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1}, 0 < x < 1$	$\mu = \frac{a}{a+b}$	$\frac{\mu(1-\mu)}{(a+b+1)}$
Chi-Square	n	$\frac{1}{2^{n/2}\Gamma(n/2)} x^{n/2-1} e^{-x/2}, x > 0$	n	$2n$
Student- t	n	$\frac{\Gamma((n+1)/2)}{\sqrt{n\pi}\Gamma(n/2)} (1 + x^2/n)^{-(n+1)/2}$	$0 (n > 1)$	$\frac{n}{n-2} (n > 2)$